



# SD-WAN — Assignment

## Question 1

Create a comparison table listing at least 5 key differences between traditional WAN and SD-WAN, focusing on how data is routed, managed, and secured.

### Answer

**Traditional WAN** generally relies on manually configured routers and fixed WAN paths, while **SD-WAN (Software-Defined Wide Area Network)** uses centralized software-based control to manage multiple WAN connections and dynamically select paths.

| Feature               | ● Traditional WAN                                                                  | ● SD-WAN                                                                                 |
|-----------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 1. Data Routing       | Traffic generally follows manually configured or relatively fixed routes           | Traffic can be dynamically steered based on application requirements and link conditions |
| 2. Management         | Devices are often configured individually                                          | Centralized controller/orchestrator provides centralized management                      |
| 3. WAN Links          | Often relies heavily on dedicated links such as MPLS                               | Can combine MPLS, broadband, fiber, 4G/5G, and other links                               |
| 4. Path Selection     | Routing decisions primarily depend on configured routing protocols and metrics     | Can consider factors such as latency, packet loss, jitter, and application requirements  |
| 5. Security           | Security may depend on separate firewalls and security appliances                  | Security policies can be centrally managed and integrated with SD-WAN solutions          |
| 6. Cloud Connectivity | Traffic may be routed through a central data center before reaching cloud services | Can provide more direct and optimized connectivity to cloud/SaaS applications            |
| 7. Scalability        | Adding and configuring new branches can require significant manual work            | Centralized management can simplify deployment and policy changes                        |
| 8. Cost               | Heavy dependence on private WAN circuits can increase costs                        | Can use lower-cost Internet links alongside or instead of some private circuits          |

### ★ Main Difference

**Traditional WAN** mainly depends on fixed, manually configured network paths, while **SD-WAN** uses centralized software control to intelligently manage and optimize multiple WAN links.

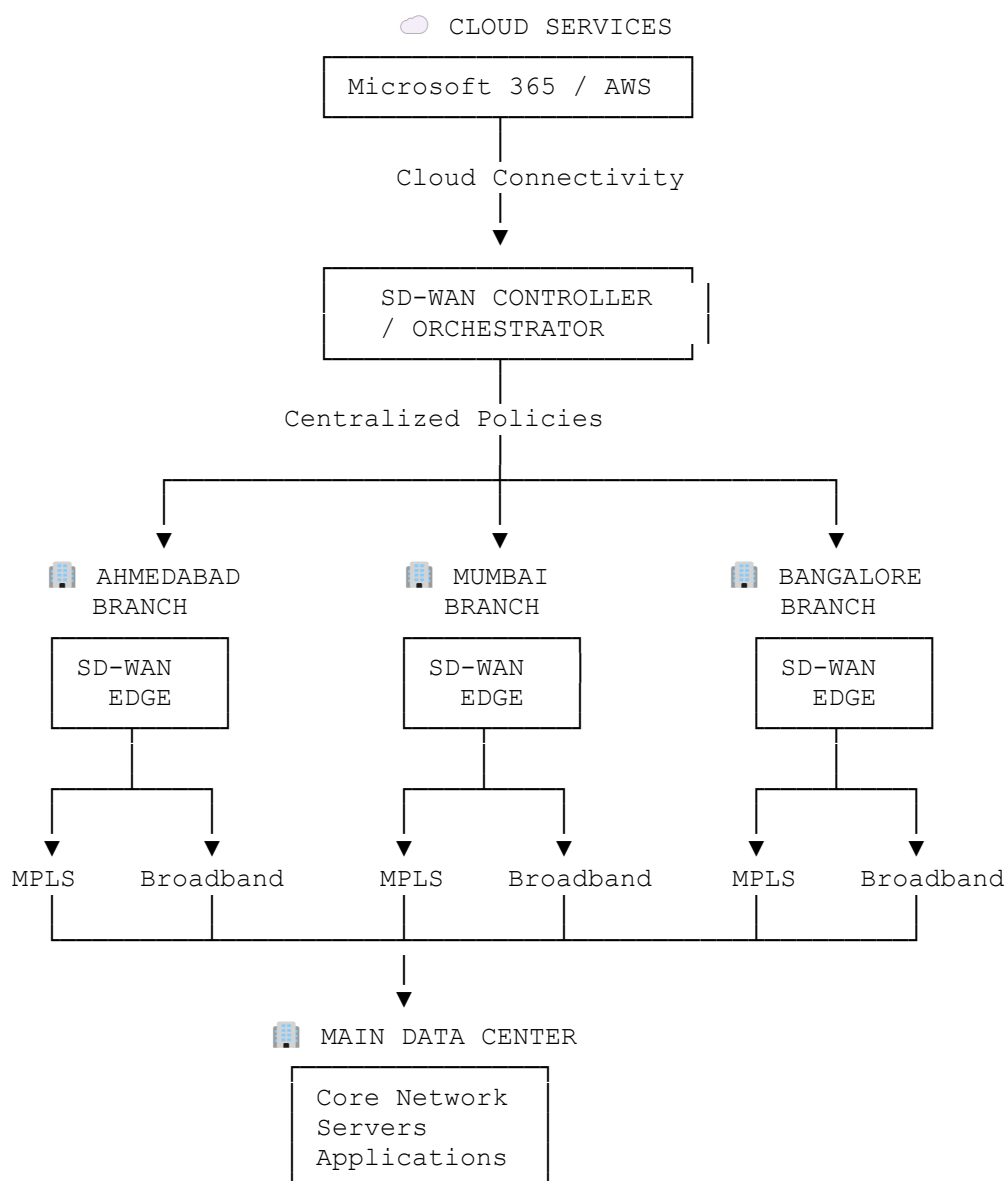
## Question 2

Draw a simple network diagram (on paper or using a tool like draw.io) showing how SD-WAN connects multiple branch offices to cloud apps (like Office 365 or AWS) and the main data center. Label the SD-WAN controllers, edges, and at least two types of WAN links (e.g., MPLS, broadband).

Answer



### SD-WAN Enterprise Network Diagram



### Main Components

### 1. SD-WAN Controller/Orchestrator

Provides centralized management and distributes policies to SD-WAN devices. It can help administrators control branch connectivity from a central location.

### 2. SD-WAN Edge

The SD-WAN edge device is installed at branch offices or other network locations. It connects to available WAN links and forwards traffic according to SD-WAN policies.

### 3. MPLS Link

MPLS can provide predictable enterprise connectivity between locations.

### 4. Broadband Link

Broadband Internet provides another WAN path and is generally less expensive than dedicated private circuits.

### 5. Cloud Services

Applications such as **Microsoft 365 and AWS** can be accessed directly through optimized WAN paths depending on the SD-WAN design.

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## Question 3

**Pick any app you use daily (like WhatsApp, Zomato, or Instagram) and write a short scenario explaining how SD-WAN could improve user experience for that app when used across multiple cities in India.**

**Hint: Think about speed, reliability, and security when users access the app from different locations.**

### Answer



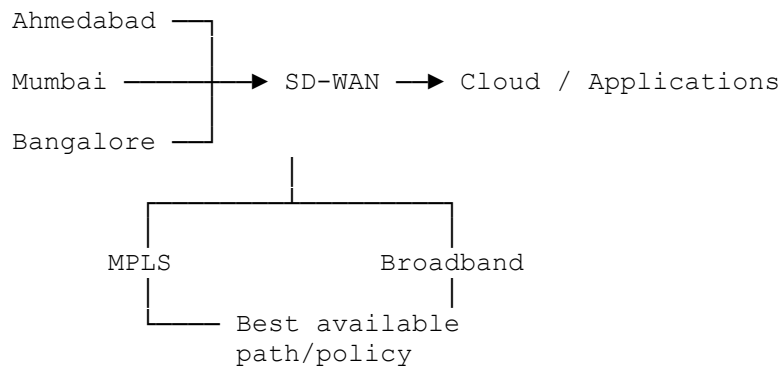
#### **Scenario: Zomato Across Multiple Indian Cities**

Consider Zomato operating offices or infrastructure across **Ahmedabad, Mumbai, Bangalore, and other cities**. Employees and backend systems may need reliable access to cloud applications, databases, and other services.

Suppose the Internet connection at the Mumbai location experiences high latency while the MPLS connection remains stable. An SD-WAN solution can monitor link conditions and select an appropriate available path according to configured policies. If another link becomes unavailable, traffic can potentially be redirected to an available connection, improving **availability and application performance**.

SD-WAN can also apply security policies to traffic moving between offices, cloud services, and other destinations. This can help the organization provide consistent network and security policies across multiple cities.

## 🚦 Simple Example



## ★ Result

- ⚡ Better application performance when suitable paths are available
- 🔄 Improved connectivity when one WAN link has problems
- 🗝️ Centralized security and traffic policies

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## Question 4

List and explain three key benefits of SD-WAN for a company with offices in Ahmedabad, Mumbai, and Bangalore. For each benefit, give a real-world example of how it would help employees or customers.

### Answer

#### 1. ⚡ Improved Application Performance

SD-WAN can monitor WAN-link conditions and steer traffic according to configured policies and application requirements.

##### Real-world example:

An employee in the **Ahmedabad office** is attending a Microsoft Teams meeting with colleagues in Mumbai and Bangalore. If one WAN path experiences high latency or packet loss, SD-WAN can use another available path according to the configured policy.

**Benefit:** Better connectivity for real-time applications such as video meetings and voice calls.

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#### 2. 🔄 Higher Reliability

SD-WAN can use multiple WAN connections such as **MPLS and broadband**. If one connection becomes unavailable, traffic can be moved to another available link according to the organization's failover policy.

**Real-world example:**

If the primary MPLS connection at the **Mumbai office** fails, the branch can use its broadband connection to maintain connectivity to important business services.

**Benefit:** Employees can continue working even when one WAN connection has a problem.

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### 3. Centralized Management and Security

SD-WAN allows network administrators to manage connectivity and security policies centrally. Instead of manually configuring every branch separately, administrators can apply consistent policies across Ahmedabad, Mumbai, and Bangalore.

**Real-world example:**

A company can create a policy requiring business-critical traffic to use approved secure paths while restricting unauthorized traffic at all three offices.

**Benefit:** More consistent network management and security across geographically separated offices.

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## Final Summary

| SD-WAN Benefit                                                                                                    | Ahmedabad                     | Mumbai                        | Bangalore                                 |
|-------------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------|-------------------------------------------|
|  <b>Performance</b>            | Optimized application traffic | Better real-time connectivity | Improved cloud access                     |
|  <b>Reliability</b>            | Backup WAN path               | MPLS + broadband failover     | Continued connectivity during link issues |
|  <b>Centralized Management</b> | Common policies               | Common policies               | Common policies                           |

### Key Takeaway

**SD-WAN uses software-based centralized control to intelligently manage multiple WAN connections, improve application connectivity, simplify management, and apply consistent network policies across different locations.**